

IC1 completes its pulse, the output goes low, turning off the relay and stopping the bell.

To prevent the doorbell from being triggered again within a short time (due to repeated pressing), a second monostable circuit built around IC2 is used. It is also configured in monostable mode and is triggered through capacitor C3 when the output of IC1 goes low after the bell has rung. IC2 then outputs a high pulse for a duration determined by resistor R6 and capacitor C5 (here it is around 50 seconds). The output of IC2 remains high for a longer duration than IC1.

This output is connected to the base of a 2N3904 transistor, which pulls the reset pin (pin 4) of IC1 low during its high state. As a result, IC1 remains disabled throughout IC2's output pulse. Even if the push-button is pressed multiple times during this period, IC1 will not retrigger, effectively preventing the bell from sounding again.

Construction and testing

An actual-size, single-sided PCB (printed circuit board) layout for the timer-controlled doorbell is shown in Fig. 3, with the component layout in Fig. 4. After assembling the components on the PCB, place the device in a suitable cabinet, ensuring that appropriate openings are provided at the rear for mounting and ventilation. This secures the unit and helps maintain operational safety.

Alternatively, if a custom PCB is not available, the circuit can be assembled on a general-purpose PCB.

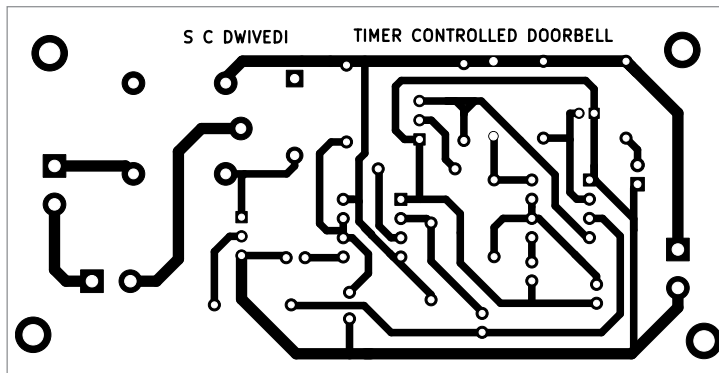


Fig. 3: Actual-size PCB

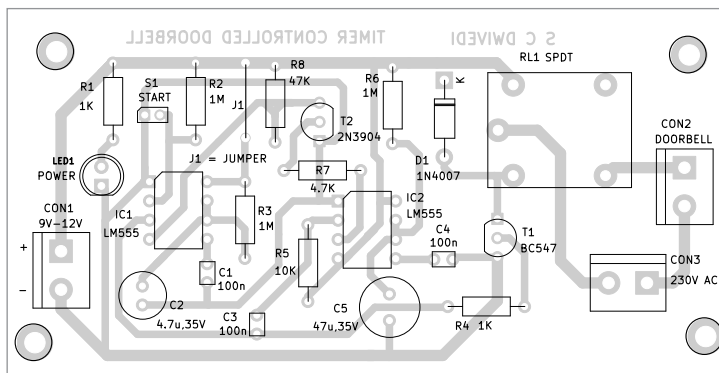


Fig. 4: Component side

Once completed, the device may be housed in a durable plastic enclosure. Ensure that the box accommodates the bell and all associated components securely, with adequate space for wiring and ventilation. A

PARTS LIST

Semiconductors:

- IC1, IC2 - LM555 timer
- T1 - BC548 NPN transistor
- T2 - 2N3904 NPN transistor
- LED1 - 5mm LED
- D1 - 1N4007 rectifier diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R4 - 1k Ω
- R2, R3, R6 - 1M Ω
- R5 - 10k Ω
- R7 - 4.7k Ω
- R8 - 47k Ω

Capacitors:

- C1, C3, C4 - 100nF ceramic disc
- C2 - 4.7 μ F, 25V electrolytic
- C5 - 47 μ F, 25V electrolytic

Miscellaneous:

- CON1 - 2-pin connector
- Battery - 9V-12V battery/adaptor
- S1 - Push-to-make switch
- RL1 - 9V/12V SPDT relay

proper enclosure keeps the device safe from dust, moisture, and accidental damage, while ensuring reliable operation. All connections should be double-checked before powering the system.

How to install

Installing this timer-controlled doorbell device is straightforward, but a careful and deliberate approach is necessary, particularly when installing it in a common area. Begin by taking the assembled PCB and connecting a pair of twin wires from the PCB to switch S1. Mount the switch near the main entrance gate. Connect the doorbell and a 230V AC power supply to the respec-

tive two-pin connectors, CON2 and CON3.

Position the assembled device in a suitable location inside the home, ensuring that the doorbell's sound is audible throughout the premises when someone presses switch S1. Use insulated wires to route the power connection safely along the mains terminals, avoiding exposed wiring. Confirm all connections are secure and test the system thoroughly before finalising the installation. This setup enhances home security and convenience with minimal complexity. **EFY**

Bonus. You can watch the video of the tutorial of this DIY project at <https://www.electronicsforu.com/videos-slideshows/live-diy-timer-controlled-doorbell>

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